

Algebraic open/closed conformal field theory possibly without vacuum: the Cardy case

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[Last page before index](#)

Contents

1	Preliminaries on conformal nets	1
1.1	Conformal nets	1
1.2	Twisted modules of conformal nets	4
1.3	The category $\text{Rep}^G(\mathcal{A})$ and the G -action	6
1.4	G -crossed categorical extensions	11
1.5	Calculus in G -crossed categorical extensions	14
	References	20

1 Preliminaries on conformal nets

1.1 Conformal nets

For each Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, we let $\mathfrak{L}(\mathcal{H}_1, \mathcal{H}_2)$ denote the set of bounded linear operators $\mathcal{H}_1 \rightarrow \mathcal{H}_2$, and abbreviate $\mathfrak{L}(\mathcal{H}, \mathcal{H})$ as $\mathfrak{L}(\mathcal{H})$. Similarly, we let $\mathcal{U}(\mathcal{H}_1, \mathcal{H}_2)$ be the set of unitary maps $\mathcal{H}_1 \rightarrow \mathcal{H}_2$ and write $\mathcal{U}(\mathcal{H}, \mathcal{H})$ as $\mathcal{U}(\mathcal{H})$. For each Hilbert space $\mathcal{H}$, we understand its inner product $\langle \cdot | \cdot \rangle$ to be linear on the right variable $|\cdot\rangle$ and antilinear on the left variable $\langle \cdot |$.

Let $\text{Diff}^+(\mathbb{S}^1)$ be the orientation-preserving diffeomorphism group of the unit circle $\mathbb{S}^1$. Let $\mathcal{I}$ be the set of (non-empty non-dense open) intervals in $\mathbb{S}^1$. If $I \in \mathcal{I}$, we let I' be the interior of $\mathbb{S}^1 \setminus I$. We let $\text{Diff}_I^+(\mathbb{S}^1)$ be the subgroup consisting of all $g \in \text{Diff}^+(\mathbb{S}^1)$ fixing points in the closure of I' .

Let $\text{PSU}(1, 1)$ be the subgroup of Möbious transforms preserving $\mathbb{S}^1$, i.e., $\text{PSU}(1, 1)$ consists of $g \in \text{Diff}^+(\mathbb{S}^1)$ of the form

$$z \in \mathbb{S}^1 \mapsto \frac{az + b}{\bar{b}z + \bar{a}}$$

where $a, b \in \mathbb{C}$, $|a|^2 - |b|^2 = 1$. Let

$$\varrho : \mathbb{R} \rightarrow \text{Diff}^+(\mathbb{S}^1) \quad \varrho(t)z = e^{it}z \quad (1.1)$$

(where $z \in \mathbb{S}^1$) be the one-parameter rotation group.

In this paper, all conformal nets are assumed to be irreducible, defined as follows.

Definition 1.1. An (irreducible) **conformal net** denotes a pair $(\mathcal{A}, \mathcal{H}_0, U_0, \Omega)$ (abbreviated to $(\mathcal{A}, \mathcal{H})$ or simply $\mathcal{A}$) where $\mathcal{H}_0$ is a separable Hilbert space, and $\mathcal{A}$ is a map sending each $I \in \mathcal{I}$ to a von Neumann algebra $\mathcal{A}(I)$ on $\mathcal{H}$ satisfying the following conditions.

- (a) **(Isotony)** If $I_1 \subset I_2 \in \mathcal{I}$, then $\mathcal{A}(I_1)$ is a von Neumann subalgebra of $\mathcal{A}(I_2)$.
- (b) **(Locality)** If $I_1, I_2 \in \mathcal{I}$ are disjoint, then $[\mathcal{A}(I_1), \mathcal{A}(I_2)] = 0$.
- (c) **(Conformal covariance)** U_0 is a strongly-continuous projectively unitary representation of $\text{Diff}^+(\mathbb{S}^1)$ on $\mathcal{H}_0$ is equipped such that the relations

$$V\mathcal{A}(I)V^* = \mathcal{A}(gI)$$

hold for any $g \in \text{Diff}^+(\mathbb{S}^1)$, any $I \in \mathcal{I}$, any unitary $V \in \mathcal{U}_0(\mathcal{H}_0)$ representing $U_0(g)$. Moreover, if $g \in \text{Diff}_I^+(\mathbb{S}^1)$ and V represents $U_0(g)$, then $V \in \mathcal{A}(I)$.

- (d) $\Omega \in \mathcal{H}_0$, called the **vacuum vector**, is the unique (up to scalar multiplication) unit vector such that $V\Omega \in \mathbb{C}\Omega$ for each $V \in \mathcal{U}(\mathcal{H})$ representing some $g \in \text{PSU}(1, 1)$. (In particular, U_0 restricts to a strongly-continuous unitary representation U_0 of $\text{PSU}(1, 1)$ on $\mathcal{H}$ satisfying $U_0(g)\Omega = \Omega$ for each $g \in \text{PSU}(1, 1)$.) Moreover, Ω is cyclic under the action of $\bigvee_{I \in \mathcal{I}} \mathcal{A}(I)$.
- (e) **(Positive energy)** The self-adjoint generator of the one-parameter unitary group $U_0 \circ \varrho$ has spectrum in $\mathbb{R}_{\geq 0}$.

Remark 1.2. A conformal net $\mathcal{A}$ satisfies the following properties, cf. [GL96] and the reference therein.

- (1) **(Additivity)** $\mathcal{A}(I) = \bigvee_{\alpha} \mathcal{A}(I_{\alpha})$ if (I_{α}) is a collection in $\mathcal{I}$ whose union is $I \in \mathcal{I}$.
- (2) **(Haag duality)** $\mathcal{A}(I)' = \mathcal{A}(I')$ for each $I \in \mathcal{I}$.
- (3) **(Reeh–Schlieder property)** $\mathcal{A}(I)\Omega$ is dense in $\mathcal{H}_0$ for each $I \in \mathcal{I}$.
- (4) **(Irreducibility)** $\bigvee_{I \in \mathcal{I}} \mathcal{A}(I) = \mathcal{L}(\mathcal{H}_0)$.
- (5) Each $\mathcal{A}(I)$ is a type III factor.

Moreover, by [MTW18], $\mathcal{A}$ satisfies the **split property**, which means that for any $I_1, I_2 \in \mathcal{J}$ with disjoint closures, there is a unitary map $\Phi : \mathcal{H}_0 \rightarrow \mathcal{H}_0 \otimes \mathcal{H}_0$ such that

$$\Phi \mathcal{A}(I_1) \Phi^{-1} = \mathcal{A}(I_1) \otimes \mathbb{C} \quad \Phi \mathcal{A}(I_2) \Phi^{-1} = \mathbb{C} \otimes \mathcal{A}(I_2)$$

By the Haag duality (applied to $I = I_1$ and $J = I'_2$), this is equivalent to the statement that for each $I, J \in \mathcal{J}$ with $I \subseteq J$, there is a type I factor between $\mathcal{A}(I)$ and $\mathcal{A}(J)$.

Next, we review some useful extensions of $\text{Diff}^+(\mathbb{S}^1)$ and $\text{Diff}_I^+(\mathbb{S}^1)$. Let

$$\mathcal{G} = \widetilde{\text{Diff}^+(\mathbb{S}^1)} \rightarrow \text{Diff}^+(\mathbb{S}^1) \quad (1.2)$$

be the universal cover of $\text{Diff}^+(\mathbb{S}^1)$. Viewing $\mathbb{S}^1$ as $\mathbb{R}/2\pi\mathbb{Z}$, the group $\mathcal{G}$ consists of smooth functions $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for each $x \in \mathbb{R}$ that

$$F(x + 2\pi) = F(x) + 2\pi \quad F'(x) > 0 \quad (1.3)$$

For each $I \in \mathcal{J}$, let $\mathcal{G}(I)$ be the identity component of the preimage of $\text{Diff}_I^+(\mathbb{S}^1)$ under the map (1.2). In other words, $\mathcal{G}(I)$ consists of smooth functions $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$ satisfying (1.3) and fixing the points outside the preimage of I under the covering map $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$.

Lift the rotation group (1.1) to a one-parameter group

$$\varrho : \mathbb{R} \rightarrow \mathcal{G}$$

Thus, for each $t \in \mathbb{R}$, the function $\varrho(t) : \mathbb{R} \rightarrow \mathbb{R}$ sends x to $x + t$.

The **central extension of $\mathcal{G}$ associated to $\mathcal{A}$** is defined to be the topological group

$$\mathcal{G}_{\mathcal{A}} = \{(g, V) \in \mathcal{G} \times \mathcal{U}(\mathcal{H}_0) : V \text{ represents } U_0(g)\}$$

or more precisely, the surjective group homomorphism $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$ defined by the projection $\mathcal{G} \times \mathcal{U}(\mathcal{H}_0) \rightarrow \mathcal{G}$. The kernel of this homomorphism is $1 \times \mathbb{S}^1 \simeq \mathbb{S}^1$.

Remark 1.3. The topological group $\mathcal{G}_{\mathcal{A}}$ depends only on the central charge of $\mathcal{A}$. Indeed, the net $\text{Vir}_{\mathcal{A}} : I \in \mathcal{J} \mapsto U_0(\text{Diff}_I^+(\mathbb{S}^1))''$ acting on $\overline{\text{Vir}_{\mathcal{A}}\Omega}$ is a conformal subnet of $\mathcal{A}$. Therefore, by the irreducibility in Rem. 1.2, $\overline{\text{Vir}_{\mathcal{A}}\Omega}$ is irreducible as a strongly-continuous projectively-unitary positive-energy representation of $\text{Diff}^+(\mathbb{S}^1)$, and hence is integrated from the unitary module $L(c, 0)$ (for some $c \geq 0$, called the **central charge** of $\mathcal{A}$) of the Virasoro algebra, cf. [Car04, Thm. A.1]. Thus $\text{Vir}_{\mathcal{A}}$ can be identified with the Virasoro net Vir_c with central charge c . Set

$$\mathcal{G}_c := \mathcal{G}_{\text{Vir}_c}$$

Viewing $L(c, 0)$ as the Hilbert space for Vir_c and identifying it with $\overline{\text{Vir}_{\mathcal{A}}\Omega}$, the map

$$\mathcal{G}_{\mathcal{A}} \xrightarrow{\simeq} \mathcal{G}_c \quad (g, V) \mapsto (g, V|_{L(c,0)})$$

defines an isomorphism of topological groups $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$. We will freely identify $\mathcal{G}_{\mathcal{A}}$ and $\mathcal{G}_c$ throughout the rest of this paper.

Definition 1.4. If $\tilde{g} = (g, V) \in \mathcal{G}_{\mathcal{A}}$, we write $V \in \mathcal{U}(\mathcal{H}_0)$ as $U_0(\tilde{g})$, that is,

$$\tilde{g} = (g, U_0(\tilde{g}))$$

Then U_0 is a strongly-continuous unitary representation of $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ on $\mathcal{H}_0$.

For each $I \in \mathcal{J}$, we let $\mathcal{G}_{\mathcal{A}}(I)$ be the preimage of $\mathcal{G}(I)$ under the map $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$. In particular, we let $\mathcal{G}_c(I) = \mathcal{G}_{\text{Vir}_c}(I)$. By the conformal covariance in Def. 1.1, we have

$$U_0(\mathcal{G}_{\mathcal{A}}(I)) \subset \mathcal{A}(I)$$

Remark 1.5. From the definition of conformal nets, we have a unitary representation U_0 of $\text{PSU}(1, 1)$ rather than merely a projectively unitary one. Thus, we have a one parameter unitary group

$$\varrho_{\mathcal{A}} : \mathbb{R} \rightarrow \mathcal{G}_{\mathcal{A}} \quad \varrho_{\mathcal{A}}(t) = (\varrho(t), U_0(\varrho(t)))$$

called the **rotation group in $\mathcal{G}_{\mathcal{A}}$** . We let ϱ_c denote the rotation group ϱ_{Vir_c} of Vir_c . The isomorphism $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ clearly identifies $\varrho_{\mathcal{A}}$ with ϱ_c .

1.2 Twisted modules of conformal nets

Fix a conformal net $\mathcal{A}$. An **automorphism** of $\mathcal{A}$ denotes a unitary operator $\Phi \in \mathcal{U}(\mathcal{H}_0)$ fixing Ω and satisfying $\Phi \mathcal{A}(I) \Phi^{-1} = \mathcal{A}(I)$ for each $I \in \mathcal{J}$.

We fix a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$ where G is a discrete group, and $\text{Aut}(\mathcal{A})$ is the group of automorphisms of $\mathcal{A}$. We view each $\phi \in G$ as a unitary operator on $\mathcal{H}_0$. Then

$$\phi \mathcal{A}(I) \phi^{-1} = \mathcal{A}(I)$$

Hence G gives rise to a group of automorphisms of each $\mathcal{A}(I)$.

For each $I \in \mathcal{J}$, an **arg function** denotes a continuous function $\arg_I : I \rightarrow \mathbb{R}$ such that $z = e^{i \arg_I(z)}$ for all $z \in I$. An **arg-valued interval** denotes a pair

$$\tilde{I} = (I, \arg_I)$$

where $I \in \mathcal{J}$, and $\arg_I$ is an arg function of I . The set of arg-valued intervals is denoted by $\tilde{\mathcal{J}}$.

Equivalently, $\tilde{I}$ is a connected component of the preimage of I under the covering map $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z} \simeq \mathbb{S}^1$. Therefore, since the universal cover $\mathcal{G}$ acts on $\mathbb{R}$ (cf. (1.3)), it also acts on $\tilde{\mathcal{J}}$. Through the quotient homomorphism $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$, the group $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ also acts on $\mathbb{S}^1$, $\mathbb{R}$, and $\tilde{\mathcal{J}}$.

If $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$, we write $\tilde{I} \subset \tilde{J}$ if $I \subset J$ and $\arg_J|_I = \arg_I$. We say that $\tilde{I}$ and $\tilde{J}$ are **disjoint** if $I \cap J = \emptyset$. (Note that this is not the same as saying that $\tilde{I}$ and $\tilde{J}$ are disjoint as subsets of $\mathbb{R}$.) We say that $\tilde{J}$ is **clockwise** to $\tilde{I}$ (equivalently, that $\tilde{I}$ is **anticlockwise** to $\tilde{J}$) if

$$\arg \zeta < \arg z < \arg \zeta + 2\pi \quad \text{for each } z \in I, \zeta \in J$$

Definition 1.6. The **clockwise complement** $\tilde{I}'$ (resp. the **anticlockwise complement** $\tilde{I}''$) of $\tilde{I} \in \tilde{\mathcal{J}}$ is defined to be the interval I' together with the arg function so that $\tilde{I}'$ is clockwise to $\tilde{I}$ (resp. (resp. $\tilde{I}''$ is clockwise to $\tilde{I}$)).

Definition 1.7. Let $\phi \in G$. A ϕ -**twisted $\mathcal{A}$ -module** denotes a pair $(\mathcal{H}_i, \pi_i)$ (or simply $\mathcal{H}_i$) where $\mathcal{H}_i$ is a separable Hilbert space, and π_i associates to each $\tilde{I} \in \tilde{\mathcal{J}}$ a normal unitary representation $\pi_{i,\tilde{I}}$ of $\mathcal{A}(I)$ on $\mathcal{H}$ satisfying the following properties.

- (a) If $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ and $\tilde{I} \subset \tilde{J}$, then $\pi_{i,\tilde{I}}(x) = \pi_{i,\tilde{J}}(x)$ for each $x \in \mathcal{A}(I)$.
- (b) For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$, we have

$$\pi_{i,\varrho(2\pi)\tilde{I}}(\phi x \phi^{-1}) = \pi_{i,\tilde{I}}(x) \quad (1.4)$$

If condition (b) is dropped, we say that $(\mathcal{H}_i, \pi_i)$ is a **solitonic $\mathcal{A}$ -module**.

Definition 1.8. We let $\mathbf{Rep}^\phi(\mathcal{A})$ be the W^* -category of ϕ -twisted $\mathcal{A}$ -modules. For the identity element e of G we write $\mathbf{Rep}^e(\mathcal{A})$ as $\mathbf{Rep}(\mathcal{A})$. Objects in $\mathbf{Rep}(\mathcal{A})$ are called **untwisted $\mathcal{A}$ -modules**, or simply **$\mathcal{A}$ -modules**.

Example 1.9. The action of $\mathcal{A}$ on $\mathcal{H}_0$ is clearly an untwisted $\mathcal{A}$ -module, called the **vacuum module** and is denoted by $(\mathcal{H}_0, \pi_0)$ or simply by $\mathcal{H}_0$.

Remark 1.10. Suppose that $x \in \mathcal{A}(I)$ commutes with $\phi \in G$. Then by Def. 1.7-(b), $\pi_{i,\tilde{I}}(x)$ is independent of $\arg_I$. We thus write $\pi_{i,\tilde{I}}(x)$ as $\pi_I(x)$ in this case.

For example, if $g \in \mathcal{G}_{\mathcal{A}}$, then $U_0(g)$ commutes with any $\phi \in G$; cf. [MS26, Sec. 3.5]. Hence, if $g \in \mathcal{G}_{\mathcal{A}}(I)$, we can

$$\text{write } \pi_{i,\tilde{I}}(U_0(g)) \text{ as } \pi_{i,I}(U_0(g))$$

As another example, if $(\mathcal{H}, \pi) \in \mathbf{Rep}(\mathcal{A})$, then $\pi_{i,\tilde{I}}(x)$ can be written as $\pi_{i,I}(x)$ for each $x \in \mathcal{A}(I)$. $\square$

Theorem 1.11. Any ϕ -twisted $\mathcal{A}$ -module $(\mathcal{H}_i, \pi_i)$ is **conformal covariant**, which means that there is a unique strongly-continuous unitary representation U_i of $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ on $\mathcal{H}_i$ satisfying for each $\tilde{I} \in \tilde{\mathcal{J}}, g \in \mathcal{G}_{\mathcal{A}}(I)$ the relation

$$U_i(g) = \pi_{i,I}(U_0(g)) \quad (1.5)$$

Proof. This follows from [Hen19], as explained in [Gui21, Thm. 2.2] (applied to the untwisted Vir_c -module $(\mathcal{H}_i, \pi_i|_{\text{Vir}_c})$) or in [MS26, Sec. 3.5]. In particular, the uniqueness follows from the following fact. $\square$

Lemma 1.12. Let $\mathcal{I}$ be a collection of intervals covering $\mathbb{S}^1$. Then $\bigcup_{I \in \mathcal{I}} \mathcal{G}(I)$ generates algebraically the group $\mathcal{G}$. Consequently, $\bigcup_{I \in \mathcal{I}} \mathcal{G}_{\mathcal{A}}(I)$ generates algebraically the group $\mathcal{G}_{\mathcal{A}}$.

Proof. This is due to [Hen19, Lem. 17]. $\square$

We abbreviate $U_i(g)$ to $U(g)$ or even g when no confusion arises.

Corollary 1.13. For each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, $\tilde{I} \in \tilde{\mathcal{J}}$, $x \in \mathcal{A}(I)$, and $g \in \mathcal{G}_{\mathcal{A}}$, we have

$$U_i(g)\pi_{i,\tilde{I}}(x)U_i(g)^{-1} = \pi_{i,g\tilde{I}}(U_0(g)xU_0(g)^{-1}) \quad (1.6)$$

Proof. See [MS26, Sec. 3.5]. The idea is to check (1.6) using Thm. 1.11 when g is supported in any interval whose union with I is not dense, and then conclude the general case by using Lem. 1.12. $\square$

1.3 The category $\text{Rep}^G(\mathcal{A})$ and the G -action

Fix a conformal net $\mathcal{A}$ with central charge c , a discrete group G , and a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$. Let

$$\text{Rep}^G(\mathcal{A}) = \bigoplus_{\phi \in G} \text{Rep}^{\phi}(\mathcal{A})$$

be the W^* -category whose objects are finite (orthogonal) direct sums of the form $\bigoplus_i \mathcal{H}_i$ where each $\mathcal{H}_i \in \text{Rep}^{\omega_i}(\mathcal{A})$ for some $\omega_i \in G$. Objects in $\text{Rep}^G(\mathcal{A})$ are called **G -twisted $\mathcal{A}$ -modules**.

Convention 1.14. In this paper, objects of $\text{Rep}^G(\mathcal{A})$ are denoted by symbols such as $\mathcal{H}_i, \mathcal{H}_j, \mathcal{H}_k$, etc. We abbreviate $\pi_{\tilde{I},i}$ to $\pi_{\tilde{I}}$ when no confusion arises.

Definition 1.15. If $\mathcal{H}_i, \mathcal{H}_j \in \mathcal{A}$, we let $\text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$ denote G -twisted $\mathcal{A}$ -module morphisms from $\mathcal{H}_i$ to $\mathcal{H}_j$, that is, elements $T \in \mathcal{L}(\mathcal{H}_i, \mathcal{H}_j)$ satisfying

$$T\pi_{\tilde{I},i}(x) = \pi_{\tilde{I},j}(x)T$$

for each $\tilde{I} \in \tilde{\mathcal{J}}, x \in \mathcal{A}(I)$. Elements of $\text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$ are called **(homo)morphisms of G -twisted $\mathcal{A}$ -modules** from $\mathcal{H}_i$ to $\mathcal{H}_j$.

Definition 1.16. For each $\phi \in G$ and each G -twisted $\mathcal{A}$ -module $(\mathcal{H}_i, \pi_i)$, we define a G -twisted $\mathcal{A}$ -module

$$(\mathcal{H}_{\phi i}, \pi_{\phi i})$$

abbreviated to $\mathcal{H}_{\phi i}$, where $\mathcal{H}_{\phi i}$ is an abstract Hilbert space unitarily equivalent to $\mathcal{H}_i$ via a prescribed unitary map

$$\Gamma_\phi|_{\mathcal{H}_i} : \mathcal{H}_i \xrightarrow{\sim} \mathcal{H}_{\phi i}$$

and $\pi_{\phi i}$ is defined such that for each $\tilde{I} \in \tilde{\mathcal{J}}, x \in \mathcal{A}(I)$, we have

$$\Gamma_\phi|_{\mathcal{H}_i} \circ \pi_{\tilde{I}, i}(x) = \pi_{\tilde{I}, \phi i}(\phi x \phi^{-1}) \circ \Gamma_\phi|_{\mathcal{H}_i}$$

We abbreviate $\Gamma_\phi|_{\mathcal{H}_i}$ to Γ_ϕ when no confusion arises. Then the above relation can be abbreviated to

$$\Gamma_\phi \circ \pi_{\tilde{I}}(x) = \pi_{\tilde{I}}(\phi x \phi^{-1}) \circ \Gamma_\phi \quad (1.7)$$

For each $\omega \in G$, it is clear that $\mathcal{H}_i \in \text{Rep}^\omega(\mathcal{A})$ implies $\mathcal{H}_{\phi i} \in \text{Rep}^{\phi\omega\phi^{-1}}(\mathcal{A})$.

Remark 1.17. It is obvious that the pair $(\mathcal{H}_{\phi i}, \Gamma_\phi|_{\mathcal{H}_i})$ is uniquely determined by $\mathcal{H}_i$ and ϕ up to unique unitary maps. That is, if $(\tilde{\mathcal{H}}_{\phi i}, \tilde{\Gamma}_\phi|_{\mathcal{H}_i})$ satisfies the same property, then there is a unique unitary map (indeed, a unique G -twisted $\mathcal{A}$ -module morphism) $\Psi : \tilde{\mathcal{H}}_{\phi i} \rightarrow \mathcal{H}_{\phi i}$ such that $\Gamma_\phi|_{\mathcal{H}_i} = \Psi \circ \tilde{\Gamma}_\phi|_{\mathcal{H}_i}$. We call Ψ the (unique) **unitary morphism** $(\tilde{\mathcal{H}}_{\phi i}, \tilde{\Gamma}_\phi|_{\mathcal{H}_i}) \rightarrow (\mathcal{H}_{\phi i}, \Gamma_\phi|_{\mathcal{H}_i})$.

Example 1.18. For the vacuum module $\mathcal{H}_0$, one can choose $(\tilde{\mathcal{H}}_{\phi 0}, \tilde{\Gamma}_\phi|_{\mathcal{H}_0})$ to be $(\mathcal{H}_0, \phi)$ where ϕ is understood as in $\mathcal{U}(\mathcal{H}_0)$. The unitary morphism $(\mathcal{H}_0, \phi) \rightarrow (\mathcal{H}_{\phi 0}, \Gamma_\phi|_{\mathcal{H}_0})$ is given by $\Gamma_\phi|_{\mathcal{H}_0} \circ \phi^{-1}$.

Remark 1.19. Recall from Rem. 1.10 that Ad_ϕ fixes $U_0(g)$ for each $I \in \mathcal{J}$ and $g \in \mathcal{G}_\mathcal{A}(I)$. Therefore, by (1.7), Ad_{Γ_ϕ} sends $\pi_{i, I}(U_0(g))$ to $\pi_{\phi i, I}(U_0(g))$. Thus, by the uniqueness in Thm. 1.11, we have

$$\Gamma_\phi \circ U_i(g) = U_{\phi i}(g) \circ \Gamma_\phi \quad (1.8)$$

for each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $g \in \mathcal{G}_\mathcal{A}$.

Definition 1.20. For each $\phi \in G$, define a $*$ -functor $\mathfrak{T}_\phi : \text{Rep}^G(\mathcal{A}) \rightarrow \text{Rep}^G(\mathcal{A})$ such that

$$\mathfrak{T}_\phi(\mathcal{H}_i) = \mathcal{H}_{\phi i} \quad \mathfrak{T}_\phi(S) = (\Gamma_\phi|_{\mathcal{H}_i}) \circ S \circ (\Gamma_\phi|_{\mathcal{H}_i})^{-1}$$

for each $S \in \text{Hom}_\mathcal{A}(\mathcal{H}_i, \mathcal{H}_j)$. This functor is denoted by T_ϕ in [MS26].

The following theorem is proved in [MS26, Sec. 3].

Theorem 1.21. *The W^* -category $\text{Rep}^G(\mathcal{A})$ with grading $\bigoplus_{\phi \in G} \text{Rep}^\phi(\mathcal{A})$ is canonically a G -crossed balanced W^* -tensor category defined by Connes fusion, where the action of G on $\text{Rep}^G(\mathcal{A})$ is defined by $\phi \in G \mapsto \mathfrak{T}_\phi$, and the balance ϑ is defined by*

$$\vartheta_i = \Gamma_\phi \circ U_i(\varrho_{\mathcal{A}}(2\pi)) : \mathcal{H}_i \rightarrow \mathcal{H}_{\phi i}$$

for each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$.

Convention 1.22. Let $\boxtimes$ denote the tensor product bifunctor of $\text{Rep}^G(\mathcal{A})$, and call it the **fusion product** bifunctor. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, we write $\mathcal{H}_{i \boxtimes j} := \mathcal{H}_i \boxtimes \mathcal{H}_j$. According to the definition of G -crossed tensor categories, we have

$$\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A}), \mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A}) \implies \mathcal{H}_i \boxtimes \mathcal{H}_j \in \text{Rep}^{\phi\omega}(\mathcal{A}) \quad (1.9)$$

The braiding operation is denoted by $\mathbb{B}$. More precisely, for each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, the braiding operator (which is a unitary equivalence of G -twisted $\mathcal{A}$ -modules) is denoted by

$$\mathbb{B}_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_i$$

By the coherence conditions for the W^* -tensor category $\text{Rep}^G(\mathcal{A})$, we make the identification

$$(\mathcal{H}_i \boxtimes \mathcal{H}_j) \boxtimes \mathcal{H}_k = \mathcal{H}_i \boxtimes (\mathcal{H}_j \boxtimes \mathcal{H}_k)$$

via the associator, and denote them by $\mathcal{H}_i \boxtimes \mathcal{H}_j \boxtimes \mathcal{H}_k \equiv \mathcal{H}_{i \boxtimes j \boxtimes k}$; we identify

$$\mathcal{H}_0 \boxtimes \mathcal{H}_i = \mathcal{H}_i = \mathcal{H}_i \boxtimes \mathcal{H}_0$$

using the unitors. In other words, we treat $\text{Rep}^G(\mathcal{A})$ as if it is strict. $\square$

Remark 1.23. We describe more explicitly the action of G on $\text{Rep}^G(\mathcal{A})$, which contains the following data besides the association $\phi \mapsto \mathfrak{T}_\phi$.

- (a) A unitary natural equivalence $\mathbf{A}_i : \mathcal{H}_i \simeq \mathcal{H}_{ei}$ where $e \in G$ is the group identity. This is given by the unique unitary morphism $(\mathcal{H}_i, \text{id}) \rightarrow (\mathcal{H}_{ei}, \Gamma_e|_{\mathcal{H}_i})$. So

$$\mathbf{A}_i = \Gamma_e|_{\mathcal{H}_i}$$

The naturality means the commutativity of

$$\begin{array}{ccc} \mathcal{H}_i & \xrightarrow{T} & \mathcal{H}_j \\ \mathbf{A}_i \downarrow & & \downarrow \mathbf{A}_j \\ \mathcal{H}_{ei} & \xrightarrow{\mathfrak{T}_e(T)} & \mathcal{H}_{ej} \end{array}$$

for each $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$, which is straightforward to check.

- (b) A unitary isomorphism $B_\phi : \mathcal{H}_0 \rightarrow \mathcal{H}_{\phi 0}$ for each $\phi \in G$, which is given by the unique unitary morphism $(\mathcal{H}_0, \phi) \rightarrow (\mathcal{H}_{\phi 0}, \Gamma_\phi|_{\mathcal{H}_0})$ discussed in Exp. 1.21. So

$$B_\phi = \Gamma_\phi \circ \phi^{-1}|_{\mathcal{H}_0}$$

- (c) A unitary W^* -tensor natural isomorphism $\mathfrak{T}_\phi \circ \mathfrak{T}_\psi \rightarrow \mathfrak{T}_{\phi\psi}$ for $\phi, \psi \in G$, where, for each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, the unitary equivalence

$$C_{\phi, \psi, i} : \mathcal{H}_{\phi(\psi i)} \rightarrow \mathcal{H}_{(\phi\psi)i} \quad C_{\phi, \psi, i} = \Gamma_{\phi\psi} \Gamma_\psi^{-1} \Gamma_\phi^{-1}|_{\mathcal{H}_{\phi(\psi i)}}$$

is given by the unique unitary morphism $(\mathcal{H}_{\phi(\psi i)}, \Gamma_\phi \circ \Gamma_\psi|_{\mathcal{H}_i}) \rightarrow (\mathcal{H}_{(\phi\psi)i}, \Gamma_{\phi\psi}|_{\mathcal{H}_i})$.

Unraveling the phrase “ W^* -tensor natural isomorphism” gives the following commutative diagrams, where $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$.

$$\begin{array}{ccc} \mathcal{H}_{\phi(\psi i)} & \xrightarrow{\mathfrak{T}_\phi \circ \mathfrak{T}_\psi(T)} & \mathcal{H}_{\phi(\psi j)} \\ C_{\phi, \psi, i} \downarrow & & \downarrow C_{\phi, \psi, j} \\ \mathcal{H}_{(\phi\psi)i} & \xrightarrow{\mathfrak{T}_{\phi\psi}(T)} & \mathcal{H}_{(\phi\psi)j} \end{array} \quad (1.10a)$$

$$\begin{array}{ccc} \mathcal{H}_{\phi(\psi(\omega i))} & \xrightarrow{\mathfrak{T}_\phi(C_{\psi, \omega, i})} & \mathcal{H}_{\phi((\psi\omega)i)} \\ C_{\phi, \psi, \omega i} \downarrow & & \downarrow C_{\phi, \psi\omega, i} \\ \mathcal{H}_{(\phi\psi)(\omega i)} & \xrightarrow{C_{\phi\psi, \omega, i}} & \mathcal{H}_{(\phi\psi\omega)i} \end{array} \quad (1.10b)$$

$$\begin{array}{ccc} \mathcal{H}_{e(\phi i)} & \xrightarrow{C_{e, \phi, i}} & \mathcal{H}_{\phi i} \\ & \xleftarrow{A_{\phi i}} & \\ \mathcal{H}_{\phi(ei)} & \xrightarrow{C_{\phi, e, i}} & \mathcal{H}_{\phi i} \\ & \xleftarrow{\mathfrak{T}_\phi(A_i)} & \end{array} \quad (1.10c)$$

The commutativity of these diagrams is easy to check.¹

- (d) A unitary tensorator $\mathfrak{T}_\phi(-) \boxtimes \mathfrak{T}_\phi(-) \rightarrow \mathfrak{T}_\phi(- \boxtimes -)$ for each $\phi \in G$, where, for each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, the unitary equivalence

$$D_{\phi, i, j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$$

will be described in Def. 1.31.

Unraveling the phrase “tensorator” gives the following commutative diagrams, where $S \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_{\tilde{i}})$ and $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_j, \mathcal{H}_{\tilde{j}})$, and $u_{0, \phi i} :$

¹In particular, in the first two diagrams, going from the upper-left corner to the lower-right corner along either path yields, respectively, $\Gamma_{\phi\psi} T \Gamma_\psi^{-1} \Gamma_\phi^{-1}$ and $\Gamma_{\phi\psi\omega} \Gamma_\omega^{-1} \Gamma_\psi^{-1} \Gamma_\phi^{-1}$.

$\mathcal{H}_0 \boxtimes \mathcal{H}_{\phi i} \rightarrow \mathcal{H}_{\phi i}$, $u_{0,i} : \mathcal{H}_0 \boxtimes \mathcal{H}_i \rightarrow \mathcal{H}_i$, $u_{\phi i,0} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_0 \rightarrow \mathcal{H}_{\phi i}$, and $u_{i,0} : \mathcal{H}_i \boxtimes \mathcal{H}_0 \rightarrow \mathcal{H}_i$ are the unitors.

$$\begin{array}{ccc} \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} & \xrightarrow{\mathfrak{T}_{\phi}(S) \boxtimes \mathfrak{T}_{\phi}(T)} & \mathcal{H}_{\phi \tilde{i}} \boxtimes \mathcal{H}_{\phi \tilde{j}} \\ \downarrow \mathbf{D}_{\phi,i,j} & & \downarrow \mathbf{D}_{\phi,\tilde{i},\tilde{j}} \\ \mathcal{H}_{\phi(i \boxtimes j)} & \xrightarrow{\mathfrak{T}_{\phi}(S \boxtimes T)} & \mathcal{H}_{\phi(\tilde{i} \boxtimes \tilde{j})} \end{array} \quad (1.11a)$$

$$\begin{array}{ccc} \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_{\phi k} & \xrightarrow{1 \boxtimes \mathbf{D}_{\phi,j,k}} & \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi(j \boxtimes k)} \\ \downarrow \mathbf{D}_{\phi,i,j} \boxtimes 1 & & \downarrow \mathbf{D}_{\phi,i,j \boxtimes k} \\ \mathcal{H}_{\phi(i \boxtimes j)} \boxtimes \mathcal{H}_{\phi k} & \xrightarrow{\mathbf{D}_{\phi,i \boxtimes j,k}} & \mathcal{H}_{\phi(i \boxtimes j \boxtimes k)} \end{array} \quad (1.11b)$$

$$\begin{array}{ccc} \mathcal{H}_0 \boxtimes \mathcal{H}_{\phi i} & \xrightarrow{u_{0,\phi i}} & \mathcal{H}_{\phi i} \\ \downarrow \mathbf{B}_{\phi} \boxtimes 1 & & \uparrow \mathfrak{T}_{\phi}(u_{0,i}) \\ \mathcal{H}_{\phi 0} \boxtimes \mathcal{H}_{\phi i} & \xrightarrow{\mathbf{D}_{\phi,0,i}} & \mathcal{H}_{\phi(0 \boxtimes i)} \end{array} \quad (1.11c)$$

$$\begin{array}{ccc} \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_0 & \xrightarrow{u_{\phi i,0}} & \mathcal{H}_{\phi i} \\ \downarrow 1 \boxtimes \mathbf{B}_{\phi} & & \uparrow \mathfrak{T}_{\phi}(u_{i,0}) \\ \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi 0} & \xrightarrow{\mathbf{D}_{\phi,i,0}} & \mathcal{H}_{\phi(i \boxtimes 0)} \end{array} \quad (1.11d)$$

The above compatibility conditions allow the G -action to be treated as strict, in the following sense.

Convention 1.24. Unless otherwise stated, for $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\phi, \psi \in G$, we make the identifications

$$\mathcal{H}_i = \mathcal{H}_{ei} \quad \mathcal{H}_0 = \mathcal{H}_{\phi 0} \quad \mathcal{H}_{\phi(\psi i)} = \mathcal{H}_{(\phi\psi)i} \quad (1.12a)$$

$$\mathcal{H}_{\phi i \boxtimes \phi j} = \mathcal{H}_{\phi(i \boxtimes j)} \quad (1.12b)$$

using the unitary isomorphisms **A**, **B**, **C**, **D** mentioned in Rem 1.21. Thus, for $\phi_1, \dots, \phi_n \in G$, we interpret $\mathcal{H}_{\phi_1 \dots \phi_n i}$ with parentheses inserted arbitrarily. Under these identifications, by (a,b,c) of Rem. 1.23, we have

$$\Gamma_e|_{\mathcal{H}_i} = \text{id}_{\mathcal{H}_i} \quad \Gamma_{\phi}|_{\mathcal{H}_0} = \phi \quad \Gamma_{\phi\psi}|_{\mathcal{H}_i} = \Gamma_{\phi} \circ \Gamma_{\psi}|_{\mathcal{H}_i}$$

which we abbreviate as

$$\Gamma_e = \text{id} \quad \Gamma_{\phi}|_{\mathcal{H}_0} = \phi \quad \Gamma_{\phi\psi} = \Gamma_{\phi} \circ \Gamma_{\psi} \quad (1.13)$$

In other words, the assignment $\phi \in G \mapsto \Gamma_\phi$ defines a “categorical group representation” $G \curvearrowright \text{Rep}^G(\mathcal{A})$ extending $G \curvearrowright \mathcal{H}_0$.

As a consequence of (1.13), we have

$$\Gamma_\phi^{-1} = \Gamma_{\phi^{-1}} \quad (1.14)$$

as unitary maps from each $\mathcal{H}_i$ to $\mathcal{H}_{\phi^{-1}i}$.

1.4 G -crossed categorical extensions

Let $\mathcal{A}$ be a conformal net and $G \rightarrow \text{Aut}(\mathcal{A})$ a group homomorphism. In this section, we recall the G -crossed categorical extension for $\mathcal{A}$ established in [MS26]. Recall Def. 1.6 for the clockwise and anticlockwise complements $\tilde{I}'$ and $\backslash \tilde{I}$.

Definition 1.25. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$, let

$$\begin{aligned} \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j) &= \{T \in \mathcal{L}(\mathcal{H}_i, \mathcal{H}_j) : T\pi_{i,\tilde{I}}(x) = \pi_{j,\tilde{I}}(x)T \text{ for each } x \in \mathcal{A}(I)\} \\ \mathcal{H}_j(I) &= \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j) \cdot \Omega \end{aligned}$$

where $\mathcal{H}_j(I)$ is independent of $\arg_I$. Note that the Reeh-Schlieder property implies that $\mathcal{H}_j(I)$ is dense in $\mathcal{H}_j$.

Note that Haag duality implies $\mathcal{H}_0(I) = \mathcal{A}(I)\Omega$.

Explanation. We need to explain why $\mathcal{H}_j(I)$ is independent of $\arg_I$. It suffices to show that for each $\tilde{I} \in \tilde{\mathcal{J}}$, the space $\mathcal{H}_j(I)$ defined by $\tilde{I}$ coincides with the one defined by $\varrho(2\pi)\tilde{I}$, that is,

$$\text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)\Omega = \text{Hom}_{\mathcal{A}(\varrho(2\pi)\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)\Omega$$

Assume without loss of generality that $\mathcal{H}_j \in \text{Rep}^\phi(\mathcal{A})$. Then the above relation follows from the following lemma and the fact that $\phi\Omega = \Omega$. $\square$

Lemma 1.26. Assume that $\phi \in G$ and $\mathcal{H}_j \in \text{Rep}^\phi(\mathcal{A})$. Let $T \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$. Then $T\phi \in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_j)$.

Consequently, if $\xi \in \mathcal{H}_j(I)$, and if $S \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$ and $T \in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_j)$ are the unique maps such that $S\Omega = \xi = T\Omega$, then $S = T\phi$. (The uniqueness is due to the Reeh-Schlieder property for $\mathcal{A}$.)

Proof. By (1.4), for each $x \in \mathcal{A}(I')$ we have

$$\pi_{j,\tilde{I}'}(x)T\phi = \pi_{j,\backslash \tilde{I}}(\phi x \phi^{-1})T\phi = T\phi x \phi^{-1}\phi = T\phi x$$

Thus $T\phi \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$. $\square$

Remark 1.27. By (1.7), we have

$$\Gamma_\phi \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j) \Gamma_\phi^{-1} = \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_{\phi i}, \mathcal{H}_{\phi j}) \quad (1.15)$$

In particular, noting that $\Gamma_\phi^{-1}|_{\mathcal{H}_0}$ equals ϕ^{-1} (due to (1.13)) and hence fixes Ω , we obtain

$$\Gamma_\phi(\mathcal{H}_j(I)) = \mathcal{H}_{\phi j}(I) \quad (1.16)$$

Definition 1.28. Let $A : \mathcal{P} \rightarrow \mathcal{R}$, $B : \mathcal{Q} \rightarrow \mathcal{S}$, $C : \mathcal{P} \rightarrow \mathcal{Q}$, $D : \mathcal{R} \rightarrow \mathcal{S}$ be bounded linear operators between Hilbert spaces. We say that the diagram

$$\begin{array}{ccc} \mathcal{P} & \xrightarrow{C} & \mathcal{Q} \\ A \downarrow & & \downarrow B \\ \mathcal{R} & \xrightarrow{D} & \mathcal{S} \end{array}$$

commutes adjointly if $DA = BC$ and $D^*B = AC^*$.

The following theorem continues to assume Conv. 1.22 and 1.24, except that the identification (1.12b) is imposed only at the end.

Theorem 1.29. For each $\tilde{I} \in \tilde{\mathcal{J}}$, $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, $\xi \in \mathcal{H}_i(I)$, we have bounded linear operators

$$\begin{aligned} L(\xi, \tilde{I})|_{\mathcal{H}_k} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_k, \mathcal{H}_i \boxtimes \mathcal{H}_k) \\ R(\xi, \tilde{I})|_{\mathcal{H}_k} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_k, \mathcal{H}_k \boxtimes \mathcal{H}_i) \end{aligned} \quad (1.17)$$

associated to each $\mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, called the **L operators** and the **R operators** and abbreviated to $L(\xi, \tilde{I})$, $R(\xi, \tilde{I})$ when no confusion arises, satisfying the following conditions:

- (1) (Naturality) If $\mathcal{H}_k, \mathcal{H}_{k'} \in \text{Rep}^G(\mathcal{A})$, $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_k, \mathcal{H}_{k'})$, $\xi \in \mathcal{H}_i(I)$, and $\chi \in \mathcal{H}_k$, then

$$(\mathbf{1}_i \boxtimes T)L(\xi, \tilde{I})\chi = L(\xi, \tilde{I})T\chi \quad (T \boxtimes \mathbf{1}_i)R(\xi, \tilde{I})\chi = R(\xi, \tilde{I})T\chi \quad (1.18)$$

- (2) (State-field correspondence) If $\xi \in \mathcal{H}_i(I)$, then (under the identifications $\mathcal{H}_i = \mathcal{H}_0 \boxtimes \mathcal{H}_i = \mathcal{H}_i \boxtimes \mathcal{H}_0$) we have

$$L(\xi, \tilde{I})\Omega = R(\xi, \tilde{I})\Omega = \xi \quad (1.19)$$

- (3) (Density of fusion products) The sets

$$L(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_k \quad R(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_k$$

span dense subspaces of $\mathcal{H}_i \boxtimes \mathcal{H}_k$ and $\mathcal{H}_k \boxtimes \mathcal{H}_i$, respectively.²

²Indeed, they are equal to the full space $\mathcal{H}_i \boxtimes \mathcal{H}_k$ and $\mathcal{H}_k \boxtimes \mathcal{H}_i$ respectively by the fact that $\mathcal{A}(I)$ is a type III factor.

- (4) (Locality) For any $\mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, any $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$, and any $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J)$, the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_k & \xrightarrow{R(\eta, \tilde{J})} & \mathcal{H}_k \boxtimes \mathcal{H}_j \\ L(\xi, \tilde{I}) \downarrow & & \downarrow L(\xi, \tilde{I}) \\ \mathcal{H}_i \boxtimes \mathcal{H}_k & \xrightarrow{R(\eta, \tilde{J})} & \mathcal{H}_i \boxtimes \mathcal{H}_k \boxtimes \mathcal{H}_j \end{array} \quad (1.20)$$

- (5) (Braiding) If $\phi \in G$ and $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$, the unitary isomorphism $\mathbb{B}_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_i$ satisfies

$$\mathbb{B}_{i,j} L(\xi, \tilde{I}) \eta = R(\xi, \tilde{I}) \Gamma_\phi \eta \quad (1.21)$$

whenever $\xi \in \mathcal{H}_i(I)$ and $\eta \in \mathcal{H}_j$.

- (6) (Conformal covariance) If $g \in \mathcal{G}_\mathcal{A}$ and $\xi \in \mathcal{H}_i(I)$, there exists a (necessarily unique) element denoted by $g\xi g^{-1} \in \mathcal{H}_i(gI)$ satisfying

$$L(g\xi g^{-1}, g\tilde{I}) = gL(\xi, \tilde{I})g^{-1} \quad R(g\xi g^{-1}, g\tilde{I}) = gR(\xi, \tilde{I})g^{-1} \quad (1.22)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$.

Moreover, assuming the identification (1.12b), we also have:

- (7) (G-covariance) For each $\xi \in \mathcal{H}_i(I)$ and $\phi \in G$, noting that $\Gamma_\phi \xi \in \mathcal{H}_{\phi i}(I)$ (cf. Rem. 1.27), we have

$$\Gamma_\phi L(\xi, \tilde{I}) = L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \quad \Gamma_\phi R(\xi, \tilde{I}) = R(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \quad (1.23)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ (with targets in $\mathcal{H}_{\phi(i \boxtimes j)} = \mathcal{H}_{\phi i \boxtimes \phi j}$ and $\mathcal{H}_{\phi(j \boxtimes i)} = \mathcal{H}_{\phi j \boxtimes \phi i}$, respectively).

When we wish to emphasize the dependence on $\mathcal{A}$, we write the operations L, R as $L^\mathcal{A}, R^\mathcal{A}$.

Remark 1.30. For each $g \in \mathcal{G}_\mathcal{A}$, the uniqueness of $g\xi g^{-1}$ in the conformal covariance is due to the state-field correspondence, which implies

$$g\xi g^{-1} = gL(\xi, \tilde{I})g^{-1}\Omega = gR(\xi, \tilde{I})g^{-1}\Omega \quad (1.24)$$

Applying the state-field correspondence again, we obtain

$$g\xi g^{-1} = g\xi \quad \text{if } g^{-1}\Omega = \Omega$$

For example, by the definition of conformal nets, for each $t \in \mathbb{R}$ we have $U_0(\varrho_{\mathcal{A}}(t))\Omega = \Omega$, and hence the **rotation covariance** property: We have

$$\varrho_{\mathcal{A}}(t)\mathcal{H}_i(I) = \mathcal{H}_i(\varrho(t)I) \quad (1.25)$$

and for each $\xi \in \mathcal{H}_i(I)$ we have

$$\varrho_{\mathcal{A}}(t)L(\xi, \tilde{I}) = L(\varrho_{\mathcal{A}}(t)\xi, \varrho(t)\tilde{I})\varrho_{\mathcal{A}}(t) \quad (1.26a)$$

$$\varrho_{\mathcal{A}}(t)R(\xi, \tilde{I}) = R(\varrho_{\mathcal{A}}(t)\xi, \varrho(t)\tilde{I})\varrho_{\mathcal{A}}(t) \quad (1.26b)$$

We are now ready to describe the unitary isomorphism mentioned in Rem. 1.23-(d).

Definition 1.31. The linear map $\mathbf{D}_{\phi, i, j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$ is the unique unitary map (indeed, the unique unitary isomorphism of G -twisted $\mathcal{A}$ -modules) such that

$$\Gamma_{\phi}L(\xi, \tilde{I})\eta = \mathbf{D}_{\phi, i, j} \circ L(\Gamma_{\phi}\xi, \tilde{I})\Gamma_{\phi}\eta \quad \Gamma_{\phi}R(\eta, \tilde{J})\xi = \mathbf{D}_{\phi, i, j} \circ R(\Gamma_{\phi}\eta, \tilde{J})\Gamma_{\phi}\xi \quad (1.27)$$

holds for each $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$, and each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J)$. The uniqueness follows from the density of fusion products in Thm. 1.29.

The existence of such a unitary isomorphism, as well as the compatibility conditions in Rem. 1.23(d), will be explained in Subsec. 1.5.3 and 1.5.4, respectively. The G -covariance property in Thm. 1.29 would then follow automatically.

1.5 Calculus in G -crossed categorical extensions

We continue to fix a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$ for a conformal net $\mathcal{A}$. In this section, we do not assume the identification (1.12b).

1.5.1 Conditions (2)-(4) of Thm. 1.29 imply $\text{Rep}^{\phi}(\mathcal{A}) \boxtimes \text{Rep}^{\omega}(\mathcal{A}) \subset \text{Rep}^{\phi\omega}(\mathcal{A})$

In [MS26], the fusion product of two objects in $\text{Rep}^G(\mathcal{A})$ is first constructed as a solitonic $\mathcal{A}$ -module using operators Z^+, Z^- and path continuations. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J)$, the operators

$$\begin{aligned} L(\xi, \tilde{I})|_{\mathcal{H}_0} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_i) & L(\xi, \tilde{I})|_{\mathcal{H}_j} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j, \mathcal{H}_i \boxtimes \mathcal{H}_j) \\ R(\eta, \tilde{J})|_{\mathcal{H}_0} &\in \text{Hom}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_0, \mathcal{H}_j) & R(\eta, \tilde{J})|_{\mathcal{H}_i} &\in \text{Hom}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_i, \mathcal{H}_i \boxtimes \mathcal{H}_j) \end{aligned}$$

are likewise constructed using $Z^+(\xi, \tilde{I}), Z^-(\eta, \tilde{J})$ and path continuations. The following properties are readily verified:

$$(2) \quad L(\xi, \tilde{I})\Omega = \xi \text{ and } R(\eta, \tilde{J})\Omega = \eta.$$

- (3) Both $L(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_j$ and $R(\mathcal{H}_j(J), \tilde{J})\mathcal{H}_j$ span dense subspaces of $\mathcal{H}_i \boxtimes \mathcal{H}_j$.
- (4) If $\tilde{J}$ is clockwise to $\tilde{I}$, the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_0 & \xrightarrow{R(\eta, \tilde{J})|_{\mathcal{H}_0}} & \mathcal{H}_j \\ L(\xi, \tilde{I})|_{\mathcal{H}_0} \downarrow & & \downarrow L(\xi, \tilde{I})|_{\mathcal{H}_j} \\ \mathcal{H}_i & \xrightarrow{R(\eta, \tilde{J})|_{\mathcal{H}_i}} & \mathcal{H}_i \boxtimes \mathcal{H}_j \end{array}$$

These constructions and properties are, of course, part of those stated in Thm. 1.29. In what follows, we show that they already suffice to imply (1.9).

Lemma 1.32. *If $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, and if $\tilde{J}$ is clockwise to $\tilde{I}$, then for each $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$ and each $x \in \mathcal{A}(I)$, $y \in \mathcal{A}(J)$, we have*

$$\begin{aligned} R(\xi, \tilde{I})y|_{\mathcal{H}_0} &= \pi_{\tilde{J}}(\phi^{-1}y\phi)R(\xi, \tilde{I})|_{\mathcal{H}_0} \\ L(\eta, \tilde{J})x|_{\mathcal{H}_0} &= \pi_{\tilde{I}}(\omega x \omega^{-1})L(\eta, \tilde{J})|_{\mathcal{H}_0} \end{aligned}$$

Proof. Since $\tilde{I}$ is clockwise to $\varrho(2\pi)\tilde{J}$, we have

$$R(\xi, \tilde{I})y|_{\mathcal{H}_0} = \pi_{i, \varrho(2\pi)\tilde{J}}(y)R(\xi, \tilde{I})|_{\mathcal{H}_0} = \pi_{i, \tilde{J}}(\phi^{-1}y\phi)R(\xi, \tilde{I})|_{\mathcal{H}_0}$$

where the last identity is due to the fact that $\mathcal{H}_i$ is ϕ -twisted. This proves the first identity. The second one follows from a similar argument, using the relation $\pi_{j, \varrho(-2\pi)\tilde{I}}(x) = \pi_{j, \tilde{I}}(\omega x \omega^{-1})$. $\square$

Theorem 1.33. *If $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, then the solitonic $\mathcal{A}$ -module $\mathcal{H}_i \boxtimes \mathcal{H}_j$ is $\phi\omega$ -twisted.*

Proof. Choose any $\tilde{K}^+ \in \tilde{\mathcal{J}}$, and let $\tilde{K}^- = \varrho(-2\pi)\tilde{K}^+$. Choose $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ such that $\tilde{J}$ is clockwise to $\tilde{I}$, and $\tilde{I}$ is clockwise to $\tilde{K}^+$.³

Choose any $x \in \mathcal{A}(K)$, $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$. By Lem. 1.32, we have

$$L(\xi, \tilde{I})R(\eta, \tilde{J})x\Omega = L(\xi, \tilde{I})\pi_{\tilde{K}^-}(\omega^{-1}x\omega)R(\eta, \tilde{J})\Omega = \pi_{\tilde{K}^-}(\omega^{-1}x\omega)L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega$$

and

$$R(\eta, \tilde{J})L(\xi, \tilde{I})x\Omega = R(\eta, \tilde{J})\pi_{\tilde{K}^+}(\phi x \phi^{-1})L(\xi, \tilde{I})\Omega = \pi_{\tilde{K}^+}(\phi x \phi^{-1})R(\eta, \tilde{J})L(\xi, \tilde{I})\Omega$$

Therefore, by (4), the operators $\pi_{\tilde{K}^-}(\omega^{-1}x\omega)$ and $\pi_{\tilde{K}^+}(\phi x \phi^{-1})$ are equal when acting on vectors of the form $L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega$ (which equals $L(\xi, \tilde{I})\eta$ by (2)). Thus (3) implies $\pi_{\tilde{K}^-}(\omega^{-1}x\omega) = \pi_{\tilde{K}^+}(\phi x \phi^{-1})$ and hence

$$\pi_{\tilde{K}^-}(x) = \pi_{\tilde{K}^+}(\phi \omega x \omega^{-1} \phi^{-1})$$

when acting on $\mathcal{H}_i \boxtimes \mathcal{H}_j$. Thus $\mathcal{H}_i \boxtimes \mathcal{H}_j$ is $\phi\omega$ -twisted. $\square$

³For example, if $\tilde{K}^+$ is a small neighborhood of $e^{i\pi}$, one can choose $\tilde{I}$ to be a small neighborhood of $e^{i\pi/2}$, and $\tilde{J}$ a small neighborhood of $e^{-i\pi/2}$. Then $\tilde{K}^-$ is a small neighborhood of $e^{-i\pi}$.

1.5.2 Consequences of (1)-(4) of Thm. 1.29

Proposition 1.34. *Let $\tilde{J}$ be clockwise to $\tilde{I}$, and let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$. Then*

$$L(\xi, \tilde{I})\eta = R(\eta, \tilde{J})\xi$$

Proof. By the state-field correspondence and the locality in Thm. 1.29, we have

$$L(\xi, \tilde{I})\eta = L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega = R(\eta, \tilde{J})L(\xi, \tilde{I})\Omega = R(\eta, \tilde{J})\xi$$

□

Corollary 1.35 (Isotony). *If $\tilde{I}_1 \subset \tilde{I}_2 \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I_1)$, then*

$$L(\xi, \tilde{I}_1) = L(\xi, \tilde{I}_2) \quad R(\xi, \tilde{I}_1) = R(\xi, \tilde{I}_2)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$.

Proof. Choose $\tilde{J} \in \tilde{\mathcal{J}}$ clockwise to $\tilde{I}_2$. By Prop. 1.34, for each $\eta \in \mathcal{H}_j(J)$ we have

$$L(\xi, \tilde{I}_1)\eta = R(\eta, \tilde{J})\xi = L(\xi, \tilde{I}_2)\eta$$

The second identity follows from a similar argument. □

Proposition 1.36. *Let $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$. Then, when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, we have (recalling $\mathcal{H}_0(I) = \mathcal{A}(I)\Omega$)*

$$L(x\Omega, \tilde{I}) = R(x\Omega, \tilde{I}) = \pi_{\tilde{I}}(x)$$

Proof. Choose $\tilde{J}$ clockwise to $\tilde{I}$. For each $\eta \in \mathcal{H}_j(J)$, we compute that

$$\begin{aligned} L(x\Omega, \tilde{I})\eta &= L(x\Omega, \tilde{I})R(\eta, \tilde{J})\Omega = R(\eta, \tilde{J})L(x\Omega, \tilde{I})\Omega = R(\eta, \tilde{J})x\Omega \\ &= \pi_{\tilde{I}}(x)R(\eta, \tilde{J})\Omega = \pi_{\tilde{I}}(x)\eta \end{aligned}$$

where the locality and the state-field correspondence in Thm. 1.29 have been used. This proves $L(x\Omega, \tilde{I}) = \pi_{\tilde{I}}(x)$. The second relation follows from a similar argument. □

Proposition 1.37. *If $S \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_{i'})$ and $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_j, \mathcal{H}_{j'})$ are morphisms in $\text{Rep}^G(\mathcal{A})$, then for each $\xi \in \mathcal{H}_i(I)$ and $\eta \in \mathcal{H}_j(J)$, we have*

$$(S \boxtimes T)L(\xi, \tilde{I})\eta = L(S\xi, \tilde{I})T\eta \quad (T \boxtimes S)R(\xi, \tilde{I})\eta = R(S\xi, \tilde{I})T\eta$$

Proof. We prove the second relation, as the first one can be proved in a similar way. It suffices to assume that $\eta \in \mathcal{H}_j(K)$ where $\tilde{K}$ is anticlockwise to $\tilde{I}$. By the naturality in Thm. 1.29, we have

$$(T \boxtimes S)R(\xi, \tilde{I})\eta = (\mathbf{1} \boxtimes S)(T \boxtimes \mathbf{1})R(\xi, \tilde{I})\eta = (\mathbf{1} \boxtimes S)R(\xi, \tilde{I})T\eta$$

By Prop. 1.34, the above expression equals

$$(\mathbf{1} \boxtimes S)L(T\eta, \tilde{K})\xi = L(T\eta, \tilde{K})S\xi = R(S\xi, \tilde{I})T\eta$$

□

We also note the useful fact that, as $\boxtimes$ is a $*$ -bifunctor, we have

$$(S \boxtimes T)^* = S^* \boxtimes T^* \quad (1.28)$$

Proposition 1.38. *Let $\mathcal{H}_i, \mathcal{H}_j, \mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, $\tilde{I} \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I)$. The following are true.*

(a) *If $\eta \in \mathcal{H}_j(I)$, then $L(\xi, \tilde{I})\eta \in (\mathcal{H}_i \boxtimes \mathcal{H}_j)(I)$, $R(\xi, \tilde{I})\eta \in (\mathcal{H}_j \boxtimes \mathcal{H}_i)(I)$, and*

$$\begin{aligned} L(\xi, \tilde{I})L(\eta, \tilde{I})|_{\mathcal{H}_k} &= L(L(\xi, \tilde{I})\eta, \tilde{I})|_{\mathcal{H}_k} \\ R(\xi, \tilde{I})R(\eta, \tilde{I})|_{\mathcal{H}_k} &= R(R(\xi, \tilde{I})\eta, \tilde{I})|_{\mathcal{H}_k} \end{aligned}$$

(b) *If $\psi \in (\mathcal{H}_i \boxtimes \mathcal{H}_j)(I)$ and $\eta \in (\mathcal{H}_j \boxtimes \mathcal{H}_i)(I)$, then $L(\xi, \tilde{I})^*\psi \in \mathcal{H}_j(I)$, $R(\xi, \tilde{I})^*\eta \in \mathcal{H}_i(I)$, and*

$$\begin{aligned} L(\xi, \tilde{I})^*L(\psi, \tilde{I})|_{\mathcal{H}_k} &= L(L(\xi, \tilde{I})^*\psi, \tilde{I})|_{\mathcal{H}_k} \\ R(\xi, \tilde{I})^*R(\eta, \tilde{I})|_{\mathcal{H}_k} &= R(R(\xi, \tilde{I})^*\eta, \tilde{I})|_{\mathcal{H}_k} \end{aligned}$$

Proof. We prove the first of (b); part (a) and the second of (b) follows from a similar argument. Part (a) follows as in [Gui21, Prop. 3.6].

By the state-field correspondence, we have $L(\xi, \tilde{I})^*\psi = L(\xi, \tilde{I})^*L(\psi, \tilde{I})\Omega$. This implies, noticing $L(\xi, \tilde{I})^*L(\psi, \tilde{I}) \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$, that $L(\xi, \tilde{I})^*\psi \in \mathcal{H}_j(I)$. Choose $\tilde{J}$ clockwise to $\tilde{I}$ and $\chi \in \mathcal{H}_k(J)$. Then

$$\begin{aligned} L(\xi, \tilde{I})^*L(\psi, \tilde{I})\chi &= L(\xi, \tilde{I})^*L(\psi, \tilde{I})R(\chi, \tilde{J})\Omega = R(\chi, \tilde{J})L(\xi, \tilde{I})^*L(\psi, \tilde{I})\Omega \\ &= R(\chi, \tilde{J})L(\xi, \tilde{I})^*\psi = R(\chi, \tilde{J})L(L(\xi, \tilde{I})^*\psi, \tilde{I})\Omega = L(L(\xi, \tilde{I})^*\psi, \tilde{I})R(\chi, \tilde{J})\Omega \\ &= L(L(\xi, \tilde{I})^*\psi, \tilde{I})\chi \end{aligned}$$

where the locality and the state-field correspondence in Thm. 1.29 are used. □

Lemma 1.39. *Suppose that $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$. Then for each $\tilde{I} \in \tilde{\mathcal{J}}$ and $\xi \in \mathcal{H}_i(I)$, we have*

$$L(\xi, \tilde{I})|_{\mathcal{H}_0} = R(\xi, \tilde{I})\phi|_{\mathcal{H}_0}$$

Proof. This is a direct reformulation of Lem. 1.26. □

1.5.3 G -covariance follows from (1)-(4) of Thm. 1.29

We begin the proof of the G -covariance by proving the following special case:

Lemma 1.40. *Let $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $\phi \in G$. Then for each $\tilde{I} \in \tilde{\mathcal{I}}$ and $\xi \in \mathcal{H}_i(I)$, the following identities holds as bounded linear operators $\mathcal{H}_0 \rightarrow \mathcal{H}_{\phi i}$.*

$$\Gamma_\phi L(\xi, \tilde{I})|_{\mathcal{H}_0} = L(\Gamma_\phi \xi, \tilde{I})\phi|_{\mathcal{H}_0} \quad \Gamma_\phi R(\xi, \tilde{I})|_{\mathcal{H}_0} = R(\Gamma_\phi \xi, \tilde{I})\phi|_{\mathcal{H}_0} \quad (1.29)$$

Proof. By Rem. 1.27, the operator $\Gamma_\phi L(\xi, \tilde{I})\phi^{-1}|_{\mathcal{H}_0}$ belongs to $\text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_i)$, and hence is the unique element in that set sending Ω to

$$\Gamma_\phi L(\xi, \tilde{I})\phi^{-1}\Omega = \Gamma_\phi L(\xi, \tilde{I})\Omega = \Gamma_\phi \xi$$

So it equals $L(\Gamma_\phi \xi, \tilde{I})|_{\mathcal{H}_0}$. The second identity can be proved in the same way. $\square$

Remark 1.41. In Lem. 1.40, the operators $\Gamma_\phi L(\xi, \tilde{I})|_{\mathcal{H}_0}$ and $L(\Gamma_\phi \xi, \tilde{I})\phi|_{\mathcal{H}_0}$ original have codomains $\mathcal{H}_{\phi(i \boxtimes 0)}$ and $\mathcal{H}_{\phi i \boxtimes \mathcal{H}_0}$, respectively. To ensure that the first identity in (1.29) holds, we identify $\mathcal{H}_{\phi(i \boxtimes 0)} \simeq \mathcal{H}_{\phi i} \simeq \mathcal{H}_{\phi i \boxtimes 0}$ using the unitors in (1.11d). Similarly, to obtain the second identity in (1.29), we identify $\mathcal{H}_{\phi(0 \boxtimes i)} \simeq \mathcal{H}_{\phi i} \simeq \mathcal{H}_{0 \boxtimes \phi i}$ using the unitors in (1.11c). This observation will be needed in the proofs of (1.11c) and (1.11d) in Subsec. 1.5.4.

Theorem 1.42. *For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\phi \in G$, there is a (necessarily unique) unitary isomorphism of G -twisted $\mathcal{A}$ -modules*

$$\mathbf{D}_{\phi, i, j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$$

satisfying Def. 1.31.

Proof. Step 1. It suffices to construct $\mathbf{D}_{\phi, i, j}$ satisfying the first relation of (1.27), since the second one will follow immediately from Prop. 1.34.

For each $\xi_1, \xi_2 \in \mathcal{H}_i(\tilde{I})$ and $\eta_1, \eta_2 \in \mathcal{H}_j$, by Prop. 1.38,

$$\begin{aligned} \langle L(\Gamma_\phi \xi_2, \tilde{I})\Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_1, \tilde{I})\Gamma_\phi \eta_1 \rangle &= \langle \Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_2, \tilde{I})^* L(\Gamma_\phi \xi_1, \tilde{I})\Gamma_\phi \eta_1 \rangle \\ &= \langle \Gamma_\phi \eta_2 | L(L(\Gamma_\phi \xi_2, \tilde{I})^* \Gamma_\phi \xi_1, \tilde{I})\Gamma_\phi \eta_1 \rangle \end{aligned}$$

Set $x = L(\Gamma_\phi \xi_2, \tilde{I})^* L(\Gamma_\phi \xi_1, \tilde{I})|_{\mathcal{H}_0}$, which belongs to $\mathcal{A}(I)$. Then the above expression equals

$$\langle \eta_2 | \Gamma_\phi^{-1} L(x\Omega, \tilde{I})\Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \Gamma_\phi^{-1} \pi_{\tilde{I}}(x)\Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(\phi^{-1}x\phi)\eta_1 \rangle$$

where Prop. 1.36 and (1.7) are used. By Lem. 1.40,

$$\phi^{-1}x\phi = (L(\Gamma_\phi \xi_2, \tilde{I})\phi|_{\mathcal{H}_0})^* (L(\Gamma_\phi \xi_1, \tilde{I})\phi|_{\mathcal{H}_0}) = L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})|_{\mathcal{H}_0}$$

Thus

$$\langle L(\Gamma_\phi \xi_2, \tilde{I}) \Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_1, \tilde{I}) \Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I}) |_{\mathcal{H}_0}) \eta_1 \rangle$$

A similar (and indeed simpler) computation shows

$$\langle L(\xi_2, \tilde{I}) \eta_2 | L(\xi_1, \tilde{I}) \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I}) |_{\mathcal{H}_0}) \eta_1 \rangle$$

Thus, by the density of fusion products in Thm. 1.29, there is a unitary map

$$\mathbf{D}_{\phi, i, j}^{\tilde{I}} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)} \quad L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta \mapsto \Gamma_\phi L(\xi, \tilde{I}) \eta$$

for each $\xi \in \mathcal{H}_i(\tilde{I}), \eta \in \mathcal{H}_j$.

Step 2. If $\tilde{I}_1, \tilde{I}_2 \in \tilde{\mathcal{J}}$ are close in the sense that there exists $\tilde{I}_0 \in \tilde{\mathcal{J}}$ satisfying $\tilde{I}_0 \subset \tilde{I}_1$ and $\tilde{I}_0 \subset \tilde{I}_2$, then $\mathbf{D}_{\phi, i, j}^{\tilde{I}_1}$ and $\mathbf{D}_{\phi, i, j}^{\tilde{I}_2}$ agree on vectors of the form $L(\mathcal{H}_{\phi i}(I_0), \tilde{I}_0) \mathcal{H}_{\phi j}$, and hence agree on $\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j}$ by the density of fusion products. If $\tilde{I}_1$ and $\tilde{I}_2$ are not close, one can connect them by a chain of arg-valued intervals such that each interval in the chain is close to the preceding one. It follows that $\mathbf{D}_{\phi, i, j}^{\tilde{I}_1}$ and $\mathbf{D}_{\phi, i, j}^{\tilde{I}_2}$ also agree in this general case. We have thus proved that $\mathbf{D}_{\phi, i, j}^{\tilde{I}}$ is independent of the choice of $\tilde{I}$, and hence can be denoted by $\mathbf{D}_{\phi, i, j}$.

Step 3. For each $\tilde{J} \in \tilde{\mathcal{J}}$ and $y \in \mathcal{A}(y)$, and for each $\tilde{I}$ anticlockwise to $\tilde{J}$ and each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$, we compute using (1.7) that

$$\begin{aligned} \mathbf{D}_{\phi, i, j} \pi_{\tilde{J}}(y) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta &= \mathbf{D}_{\phi, i, j} L(\Gamma_\phi \xi, \tilde{I}) \pi_{\tilde{J}}(y) \Gamma_\phi \eta \\ &= \mathbf{D}_{\phi, i, j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \pi_{\tilde{J}}(\phi^{-1} y \phi) \eta = \Gamma_\phi L(\xi, \tilde{I}) \pi_{\tilde{J}}(\phi^{-1} y \phi) \eta \\ &= \Gamma_\phi \pi_{\tilde{J}}(\phi^{-1} y \phi) L(\xi, \tilde{I}) \eta = \pi_{\tilde{J}}(y) \Gamma_\phi L(\xi, \tilde{I}) \eta = \pi_{\tilde{J}}(y) \mathbf{D}_{\phi, i, j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta \end{aligned}$$

This proves that $\mathbf{D}_{\phi, i, j}$ is a morphism of G -twisted $\mathcal{A}$ -modules. $\square$

1.5.4 The compatibility conditions in Rem. 1.23-(d) follow from (1)-(4) of Thm. 1.29

Proof of (1.11a). Choose any $\tilde{I} \in \tilde{\mathcal{J}}, \xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$, we have

$$\begin{aligned} (\mathfrak{T}_\phi(S) \boxtimes \mathfrak{T}_\phi(T)) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta &= (\Gamma_\phi S \Gamma_\phi^{-1} \boxtimes \Gamma_\phi T \Gamma_\phi^{-1}) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta \\ &= L(\Gamma_\phi S \xi, \tilde{I}) \Gamma_\phi T \eta \end{aligned}$$

where Prop. 1.37 is used. Thus, by Def. 1.31,

$$\mathbf{D}_{\phi, i, j}^{\tilde{I}} (\mathfrak{T}_\phi(S) \boxtimes \mathfrak{T}_\phi(T)) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta = \Gamma_\phi L(S \xi, \tilde{I}) T \eta$$

Def. 1.31 and Prop. 1.37 also imply

$$\begin{aligned} & \mathfrak{T}_\phi(S \boxtimes T) \mathbf{D}_{\phi, i, j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta = \Gamma_\phi(S \boxtimes T) \Gamma_\phi^{-1} \cdot \Gamma_\phi L(\xi, \tilde{I}) \eta \\ & = \Gamma_\phi L(S \xi, \tilde{I}) T \eta \end{aligned}$$

This proves (1.11a), thanks to the density of fusion products. $\square$

Proof of (1.11b). Choose any $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$. Then for each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J), \chi \in \mathcal{H}_k$,

$$\begin{aligned} & \mathbf{D}_{\phi, i, j \boxtimes k}(\mathbf{1} \boxtimes \mathbf{D}_{\phi, j, k}) L(\Gamma_\phi \xi, \tilde{I}) R(\Gamma_\phi \eta, \tilde{J}) \Gamma_\phi \chi \\ & = \mathbf{D}_{\phi, i, j \boxtimes k} L(\Gamma_\phi \xi, \tilde{I}) \mathbf{D}_{\phi, j, k} R(\Gamma_\phi \eta, \tilde{J}) \Gamma_\phi \chi \\ & = \mathbf{D}_{\phi, i, j \boxtimes k} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi R(\eta, \tilde{J}) \chi = \Gamma_\phi L(\xi, \tilde{I}) R(\eta, \tilde{J}) \chi \end{aligned}$$

where the naturality in Thm. 1.29 is used. Similarly,

$$\begin{aligned} & \mathbf{D}_{\phi, i \boxtimes j, k}(\mathbf{D}_{\phi, i, j} \boxtimes \mathbf{1}) R(\Gamma_\phi \eta, \tilde{J}) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \chi \\ & = \mathbf{D}_{\phi, i \boxtimes j, k} R(\Gamma_\phi \eta, \tilde{J}) \mathbf{D}_{\phi, i, j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \chi \\ & = \mathbf{D}_{\phi, i \boxtimes j, k} R(\Gamma_\phi \eta, \tilde{J}) \Gamma_\phi L(\xi, \tilde{I}) \chi = \Gamma_\phi R(\eta, \tilde{J}) L(\xi, \tilde{I}) \chi \end{aligned}$$

This finishes the proof, thanks to the locality and the density of fusion products in Thm. 1.29. $\square$

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YAU MATHEMATICAL SCIENCES CENTER, TSINGHUA UNIVERSITY, BEIJING, CHINA.
E-mail: binguimath@gmail.com bingui@tsinghua.edu.cn